

🔑 🔑 Assignment: Data Visualization 🔑 🔑

READ THE BIG MART SALES DATA

```
In [1]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
%matplotlib inline

import seaborn as sns
```

```
In [2]: data_BM = pd.read_csv('dataset/big_mart_sales.csv')
```

```
In [3]: data_BM.head()
```

```
Out[3]:
```

	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Identifier
0	FDA15	9.30	Low Fat	0.016047	Dairy	249.8092	OUT049
1	DRC01	5.92	Regular	0.019278	Soft Drinks	48.2692	OUT018
2	FDN15	17.50	Low Fat	0.016760	Meat	141.6180	OUT049
3	FDX07	19.20	Regular	0.000000	Fruits and Vegetables	182.0950	OUT010
4	NCD19	8.93	Low Fat	0.000000	Household	53.8614	OUT013

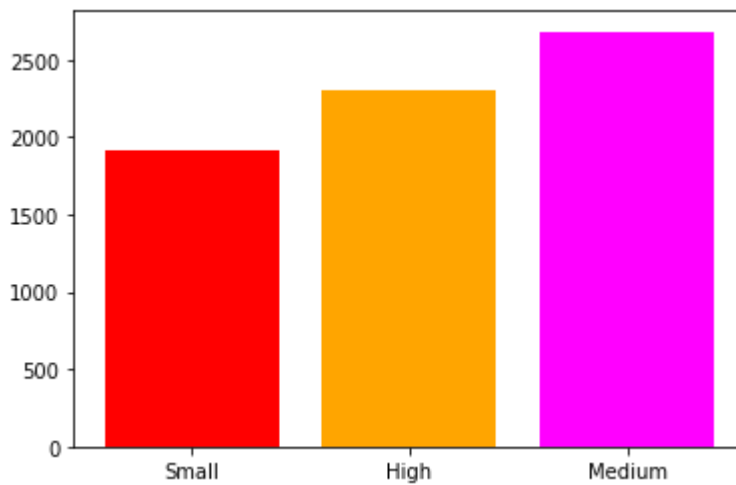
```
In [4]: # sales by outlet size
sales_by_outlet_size = data_BM.groupby('Outlet_Size').Item_Outlet_Sales.mean()

# sort by sales
sales_by_outlet_size.sort_values(inplace=True)

x = sales_by_outlet_size.index.tolist()
y = sales_by_outlet_size.values.tolist()

# set xticks
plt.xticks(labels=x, ticks=np.arange(len(x)))

plt.bar(x, y, color=['red', 'orange', 'magenta']);
```



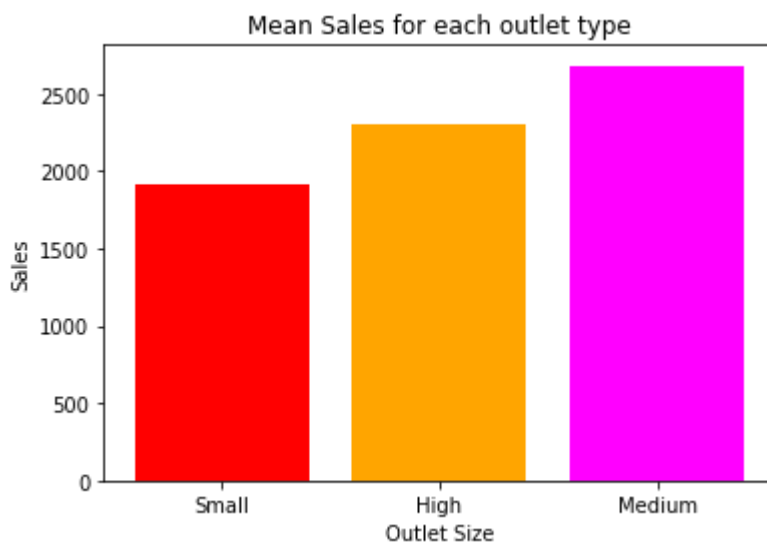
Do the following edits in the above plot

- Add the x-axis label as Outlet Size
- Add the y-axis label as Sales
- Add the plot title as Mean Sales for each outlet type

In [5]:

```
## WRITE YOUR CODE HERE
```

```
plt.xlabel('Outlet Size')  
plt.ylabel('Sales')  
plt.title('Mean Sales for each outlet type')  
plt.bar(x, y, color=['red', 'orange', 'magenta']);
```



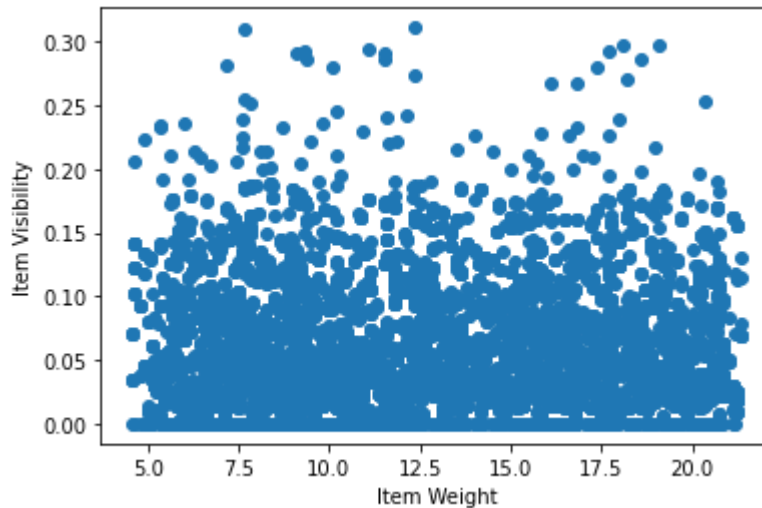
Create a scatter plot to show the relative distribution of *Item_Weight* and *Item_Visibility*.

In [6]:

```
## WRITE YOUR CODE HERE  
x = data_BM.Item_Weight
```

```
y = data_BM.Item_Visibility
plt.scatter(x, y)
plt.xlabel('Item Weight')
plt.ylabel('Item Visibility')
```

Out[6]: Text(0, 0.5, 'Item Visibility')



Create a box plot of the variable `Item_Outlet_Sales` using the seaborn.

```
In [7]: # WRITE YOUR CODE HERE
fig, ax = plt.subplots(figsize=(12, 6))
flierprops = dict(markerfacecolor='lightblue', marker='x')
plt.boxplot(data_BM.Item_Outlet_Sales, labels=['Item Outlet Sales'], flierprops=flierpr
```

